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Unit – 5 : Session – 1 : SLO – 1 | Introduction to Graphs and Terminologies

■ I. Fill in the Blanks

1. A GRAPH is a collection of vertices and edges.
2. In a graph, a connection between two nodes is called an EDGE.
3. The number of edges incident to a vertex is called the DEGREE of the vertex.
4. A graph in which edges have direction is called a DIRECTED graph.
5. A graph with values or costs assigned to each edge is called a WEIGHTED graph.

■ II. Match the Following

- a) Connection between two vertices — 2. Edge
- b) Number of edges at a vertex — 3. Degree
- c) A point where edges meet — 1. Vertex
- d) Sequence of edges joining vertices — 4. Path
- e) Path starting and ending at same vertex — 5. Cycle

■ III. Identify and Label — Cities A, B, C, D

- Vertices (cities): A, B, C, D. Edges (roads, weighted): A-B = 4 km, B-C = 1 km, C-D = 7 km, A-D = 5 km.
- Diagram: A --4-- B
- | |
- 5 1
- | |
- D --7-- C
- Degrees: $\deg(A) = 2$ (B, D); $\deg(B) = 2$ (A, C); $\deg(C) = 2$ (B, D); $\deg(D) = 2$ (C, A).

■ IV. True or False

1. In a disconnected graph, there is no path between some pairs of vertices — TRUE.
2. A path is a sequence of edges that connects a sequence of vertices — TRUE.
3. The degree of a vertex in a graph is always zero — FALSE.
4. A weighted graph has no values assigned to its edges — FALSE.
5. Two vertices connected by an edge are called adjacent vertices — TRUE.

■ V. Short Answer Questions

1. Graph: a non-linear data structure $G = (V, E)$ with a set of vertices and edges connecting pairs of
 - vertices. Example: a social network where people are vertices and friendships are edges.
2. Directed vs undirected: in a directed graph (digraph) edges have a direction ($u \rightarrow v$) and represent
 - one-way relationships; in an undirected graph edges have no direction and the relation is mutual.
3. Adjacent vertices: two vertices that are directly connected by an edge; e.g. if edge (A, B) exists,
 - A and B are adjacent (neighbours).

4. Connected graph: an undirected graph in which there is a path between every pair of vertices.
5. Applications: Google Maps / road networks (shortest path); social networks (friend suggestions);
 - also computer networks routing and web page ranking (PageRank).